

Name: _____

Class: _____

Date: _____

Mr. Rawson • IB DP Physics

SHM 3 – States, Signs and Graphs

Homework

The last set was mostly about putting numbers into formulas. This one is about knowing where the oscillator *is* and what its graphs look like, which is where a large share of the marks in this topic actually sit. Show your working, and give every answer a unit.

1) **State** A mass on a spring oscillates vertically. Take **upwards as positive**. Complete the table with the sign of each quantity, using +, – or 0.

[4]

Where the mass is	x	v	a
at its lowest point			
moving upwards through the equilibrium position			
at its highest point			
moving downwards, halfway between the top and equilibrium			

2) **Identify** Each pair below describes an oscillator at one instant. State which two are **impossible** for simple harmonic motion, and explain why in each case.

[3]

A displacement maximum, acceleration maximum C displacement maximum, velocity maximum

B displacement zero, velocity maximum D displacement zero, acceleration maximum

3) **Determine** An oscillator is released from rest at maximum displacement at $t = 0$. Its period is 1.2 s. Determine the time at which it first passes through the equilibrium position, the time at which it first reaches maximum displacement on the other side, and the number of times it passes through equilibrium during the first 3.0 s.

[4]

4) **Calculate** An object performs simple harmonic motion with a period of 0.50 s and an amplitude of 8.0 cm. Calculate its maximum acceleration, and its acceleration at the instant its displacement is 3.0 cm.

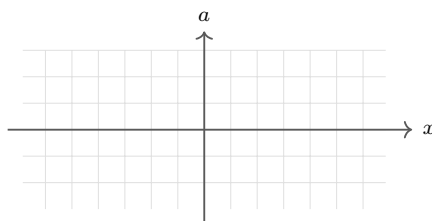
[4]

5) **Determine** For a certain oscillator, a graph of acceleration against displacement is a straight line through the origin with a gradient of -45 s^{-2} . Determine the angular frequency, the frequency and the period of the oscillation.

[3]

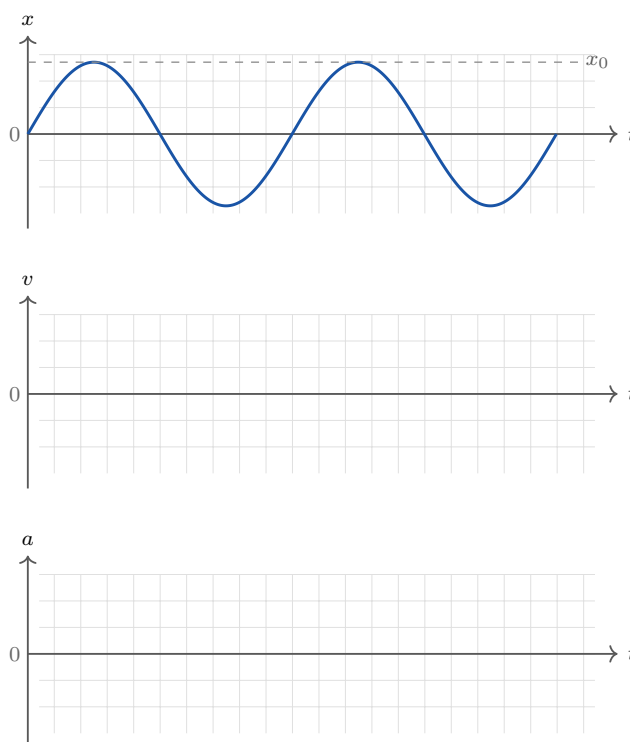
6) **Sketch** On the axes below, sketch acceleration against displacement for two oscillators, **P** and **Q**, where **Q** has twice the angular frequency of **P**. Label each line. State what feature of a line tells you ω .

[3]



7) **Sketch** The displacement–time graph of an oscillator is given. Sketch its velocity–time and acceleration–time graphs on the axes beneath, keeping the same time scale. Then state the phase relationship between the displacement and the acceleration.

[5]



Higher level only

8) **Calculate** A mass of 0.20 kg hangs at rest on a spring of spring constant 40 N m^{-1} . It is pulled down a further 5.0 cm and released. Take $g = 9.81 \text{ m s}^{-2}$.

(a) Calculate the extension of the spring at the equilibrium position. (b) Calculate the extension at the lowest point of the oscillation. (c) Calculate the elastic potential energy stored in the spring at each of those two positions. (d) The mass loses 0.15 J of elastic potential energy on the way back up to equilibrium, but does not gain 0.15 J of kinetic energy. Explain why.

[7]
